Detector-production formulas and default operating point

NAUSICAA design note

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This note defines the fields proposed for the detector-production interface. Defaults are design inputs for the 1 mm Geant4 hit samples. The producer records the selected values, source hash, seeds and code identity in its HDF5 outputs. Pure Xe and Xe–15% He are separate electron-gas outputs of the same initial ionisation realisation.

1 Initial pairs and event location

For source hit i, with deposited energy E_i in MeV and common pair-creation energy $W = 21.9$ eV, the expected number of electron–ion pairs is

$$\mu_i = \frac{10^6 E_i}{W}. \tag{1}$$

The Fano target is $\mathrm{Var}(N_i) \simeq F\mu_i$ with $F = 0.15$ for the integer pair-count model at sufficiently large μ_i [1]. For $0 \le \mu_i < 1$, use $N_i \sim \mathrm{Bernoulli}(\mu_i)$, which preserves $\mathbb{E}[N_i] = \mu_i$ and has variance $\mu_i(1-\mu_i)$. One draw sets both the initial positive-ion and electron counts for that hit. The exact integer Fano sampler at $\mu_i \ge 1$ is to be specified and validated in the pilot.

Sample the ion drift length $L_i \sim U(0, L_{\max})$ and set the electron length $L_e = L_{\max} - L_i$, with $L_{\max} = 150$ cm by default. The source hit coordinates stay in mm; no individual carrier positions are saved.

2 Ion transport and pad frames

The thermal width for singly charged Xe_2^+ is ITACA Revisited, eq. (2.1) [2]:

$$\sigma_{i,T} = \sigma_{i,L} = 10\sqrt{2\frac{k_B}{e}\frac{T L_i}{E_d}}\ \text{mm}, \qquad \frac{k_B}{e} = 8.6173 \times 10^{-5}\ \text{V/K}, \tag{2}$$

where T is in kelvin, L_i in cm and the drift field E_d in V/cm. The implemented calculation is `src/event_analysis/production.jl:sigma_t_ion_mm`; the same thermal width is used longitudinally in the current model. At $T = 293.15$ K, $E_d = 200$ V/cm and $L_i = 150$ cm, $\sigma_i = 1.946$ mm.

The ion drift speed is $v_i = 0.1$ m/s. A $\Delta t_i = 30$ ms frame corresponds to $v_i \Delta t_i = 3$ mm; maximum drift time is 15 s. The transverse pad pitch is an independent default of 3 mm. A uniform event-level grid phase in $[0, p)$ is sampled independently in x and y.

3 Electron drift and survival

The electron-width model implemented in `config/electron_transport.toml` is

$$\sigma_T = d_T\sqrt{\frac{L_e}{P}}, \qquad \sigma_L = d_L\sqrt{\frac{L_e}{P}}, \tag{3}$$

with L_e in cm, P in bar, widths in mm, and coefficient units mm$\sqrt{\text{bar/cm}}$. The gas-model inputs and widths at the maximum drift, $L_e = 150$ cm and $P = 15$ bar, are:

Gas	d_T	d_L	σ_T (mm)	σ_L (mm)
Pure Xe	3.5	0.9	11.068	2.846
Xe–15% He	1.6	0.75	5.06	2.372

ITACA Revisited table 1 [2] gives the pure-Xe transverse coefficient $d_T = 3.5$ in these units. The remaining coefficients in this table are adopted inputs from `config/electron_transport.toml`; their numerical provenance requires a separate validation against electron transport measurements or calculations at the chosen field and gas mixture. The 15% He mixture identity is the adopted design input. At $v_e = 1$ mm/μs the maximum drift time is 1.5 ms. The electron lifetime $\tau_e = 15$ ms gives independent survival probability $p_e = \exp[-t_e/\tau_e]$. Its equivalent attenuation length is $v_e\tau_e = 15$ m. The *minimum* survival, at maximum drift, is 0.905. Electron attachment changes only the electron branch. A $\Delta t_e = 2$ μs bucket corresponds to 2 mm.

4 EL light, fluctuations and optical collection

For a hole depth h in cm, the gas-specific fitted mean EL yield is

$$Y_g = a_g \max(E/P - x_{0,g}, 0)\, Ph \quad \text{photons per entering electron,} \tag{4}$$

with E/P in kV/cm/bar and a_g in photons/electron/kV [3, table 1]. The pure-Xe parameters are $(a, x_0) = (137, 0.73)$; the Xe–15% He parameters are $(139, 0.80)$. At $E/P = 3$, $P = 15$ bar and $h = 0.5$ cm, $Y_{\text{Xe}} = 2{,}332.4$ and $Y_{\text{XeHe}} = 2{,}293.5$ photons/electron.

The EL fluctuation parameter is $J(E/P) = \text{Var}(N_{\text{EL}} \mid 1\ e^-)/Y_g$ as defined by Oliveira et al. [1, eq. (4.6)]. The design uses the same $J(E/P)$ curve for both gas models as an adopted approximation. The user supplied the following approximate readings of the pure-Xe curve in Figure 6 of that paper:

E/P (kV/cm/bar)	J
1.5	0.10
2.0	0.05
3.0	0.01

The default operating point uses $J = 0.01$. At fields between these three anchors, the production form requires an explicit J value; it does not interpolate or extrapolate the plotted curve.

For isotropic VUV emission uniformly along the axis of a cylindrical hole, absorbing walls and a SiPM covering its exit, the direct geometric acceptance is

$$A(r, h) = \frac{h + r - \sqrt{h^2 + r^2}}{2h}. \tag{5}$$

For radius $r = 2.5$ mm and depth $h = 5$ mm, $A = 0.19098$. Conditional on emitted photons, the incident count is binomial with success probability A. The corresponding mean incident photons per entering electron are 445.4 (pure Xe) and 438 (Xe–He). SiPM PDE is applied in the subsequent response stage.

5 Electrode potentials and grid assignment

Choose the cathode as the zero of potential. For drift field E_d and TPC length $L_{\max}$,

$$V_{\text{gate}} = E_d L_{\max}, \qquad V_{\text{anode}} = V_{\text{gate}} + 10^3 (E/P) Ph. \tag{6}$$

The defaults give $V_{\mathrm{gate}} = 30$ kV and $V_{\mathrm{anode}} = 52.5$ kV. The electron-hole pitch is 10 mm and the hole diameter and depth are 5 mm. The source-energy-weighted transverse centre and minimum source z define the ion-grid reference. For either transverse grid, one event-level phase is uniform over a pitch; the physical cell index is obtained by the floor of the position relative to that reference and phase, divided by the pitch.

References

[1] C. A. B. Oliveira, M. Sorel, J. Martin-Albo, J. J. Gomez-Cadenas, A. L. Ferreira, and J. F. C. A. Veloso. Energy Resolution studies for NEXT. *JINST*, 6:P05007, 2011.

[2] J. J. Gómez-Cadenas, L. Arazi, G. Martínez-Lema, J. Renner, S. R. Soleti, and S. Torelli. ITACA revisited: Ion tracking apparatus with CMOS ASICs, 2026. Version 1. Its original IFG voltages are superseded by the design revision used in this note.

[3] A. F. M. Fernandes et al. Low-diffusion Xe-He gas mixtures for rare-event detection: Electroluminescence Yield. *JHEP*, 04:034, 2020.